Grade: Check Plus

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ASSIGNMENT 2: EXPECTED VALUES AND OPTIONS

1. Compute the mean and standard deviation of the stock price.

Price (X)	**Probability f(x)**	**Mean (μ)** *Sum of x*f(x)*	**Variance** *Sum* of $(x - \mu)^2 f(x)$
20	.05	1.00	1.46
21	.06	1.26	1.17
22	.07	1.54	0.81
23	.08	1.84	0.46
24	.09	2.16	0.18
25	.12	3.00	0.02
26	.15	3.90	0.05
27	.13	3.51	0.33
28	.10	2.80	0.67
29	.10	2.90	1.29
30	.05	1.50	1.05
	1.00	**25.41**	7.49

Mean (μ) = 20(0.05) + 21(0.06) + 22(0.07) + 23(0.08) + 24(0.09) + 25(0.12) + 26(0.15) + 27(0.13) + 28(0.10) + 29(0.10) + 30(0.05)
= 1.00 + 1.26 + 1.54 + 1.84 + 2.16 + 3.00 + 3.90 + 3.51 + 2.80 + 2.90 + 1.50
= **25.41**

x Q1: Rounding in intermediate steps.

Variance = $(20-25.41)^2(0.05) + (21-25.41)^2(0.06) + (22-25.41)^2(0.07) + (23-25.41)^2(0.08) + (24-25.41)^2(0.09) + (25-25.41)^2(0.12) + (26-25.41)^2(0.15) + (27-25.41)^2(0.13) + (28-25.41)^2(0.10) + (29-25.41)^2(0.10) + (30-25.41)^2(0.05)$
= 7.49

Standard Deviation (SD) = √variance = √7.49 = **2.74**
Therefore, the expected stock price next year is about $25.41, with typical variation of about $2.74.

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2(a). What is the probability that the stock price will fall within 1 standard deviation of the mean one year from now?

P(Mean - SD ≤ X ≤ Mean + SD) =
P(25.41 - 2.74 ≤ X ≤ 25.41 + 2.74) =
P(22.67 ≤ X ≤ 28.15)
✓ Q2a
We take the stock prices that are between 22.67 and 28.15, which are 23, 24, 25, 26, 27 and 28. Therefore, P = 0.08 + 0.09 + 0.12 + 0.15 + 0.13 + 0.10 = **0.67**

The probability that the stock price will fall within 1 standard deviation of the mean one year from now is 0.67 or 67%.

2(b). What is the probability that the stock price will fall within 2 standard deviations of the mean one year from now?

P(Mean - 2xSD ≤ X ≤ Mean + 2xSD) =
P(25.41 - 5.48 ≤ X ≤ 25.41 + 5.48) =
✓ Q2b
P(19.93 ≤ X ≤ 30.89) - this means that all the stock prices (20-30) are possible as they all fall in this interval. Therefore, P = **1.00**

The probability that the stock price will fall within 2 standard deviations of the mean one year from now is 1.00 or 100%

3. Derive the distribution of the option value, ignoring interest and commissions.

Price (X)	**Probability f(x)**	**Option Value** *X - $25*
20	.05	0
21	.06	0
22	.07	0
23	.08	0
24	.09	0
25	.12	0

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26	.15	1
27	.13	2
28	.10	3
29	.10	4
30	.05	5
	1.00	

Option value 0 = 0.05+0.06+0.07+0.08+0.09+0.12 = **0.47**
Option value 1 = **0.15**
Option value 2 = **0.13**
Option value 3 = **0.10**
Option value 4 = **0.10**
Option value 5 = **0.05**

The probability distribution of the option value is shown in the table below:

Option Value (V)	**Probability g(v)**
0	0.47
1	0.15
2	0.13
3	0.10
4	0.10
5	0.05
	1.00

✓ Q3

4. What is the probability that the option will have positive value?

✓ Q4

The probability that the option will have positive value = 1 - 0.47 = **0.53.**
The option is worthless in 47% of outcomes and positive in the remaining 53% (0.53).

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5. Compute the mean and standard deviation of the value of the option.

Option Value (v)	**Probability g(v)**	**Mean (μ)** *Sum of v*g(v)*	**Variance** *Sum* of $(v - \mu)^2 g(v)$
0	0.47	0	0.87
1	0.15	0.15	0.02
2	0.13	0.26	0.05
3	0.10	0.30	0.27
4	0.10	0.40	0.70
5	0.05	0.25	0.66
	1.00	**1.36**	2.57

Mean (μ) = 1(0.15) + 2(0.13) + 3(0.10) + 4(0.10) + 5(0.05)
= 0.15 + 0.26 + 0.30 + 0.40 + 0.25
= **1.36**

Variance = $(0-1.36)^2(0.47) + (1-1.36)^2(0.15) + (2-1.36)^2(0.13) + (3-1.36)^2(0.10) + (4-1.36)^2(0.10) + (5-1.36)^2(0.05) = 2.57$

✓ Q5

Standard Deviation (SD) = √variance = √2.57 = **1.60**

On average the option is worth about $1.36, with typical variation of about $1.60.